

A Project Report

on

AI TOOLS USAGE

Submitted in partial fulfillment of requirements for the award of the course

of

AMI1201 – DATA EXPLORATION AND VISUALIZATION

Under the guidance of

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Submitted By

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**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE
LEARNING**

M.KUMARASAMY COLLEGE OF ENGINEERING
(Autonomous)

KARUR – 639 113

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M. KUMARASAMY COLLEGE OF ENGINEERING
(Autonomous Institution affiliated to Anna University, Chennai)

KARUR – 639 113

BONAFIDE CERTIFICATE

Certified that this project report on “**AI TOOLS USAGE**” is bonafide work of **VIVEKA E(927624BAM061)** who carried out the project work during the academic year 2025 - 2026 under my supervision.

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

VISION OF THE INSTITUTION

To emerge as a leader among the top institutions in the field of technical education

MISSION OF THE INSTITUTION

- Produce smart technocrats with empirical knowledge who can surmount the global challenges
- Create a diverse, fully-engaged, learner-centric campus environment to provide quality education to the students
- Maintain mutually beneficial partnerships with our alumni, industry, and Professional associations

VISION OF THE DEPARTMENT

To become a renowned hub for AIML technologies to producing highly talented globally recognizable technocrats to meet industrial needs and societal expectation.

MISSION OF THE DEPARTMENT

M1: To impart advanced education in AI and Machine Learning, built upon a foundation in Computer Science and Engineering.

M2: To foster Experiential learning equips students with engineering skills to tackle real-world problems.

M3: To promote collaborative innovation in AI, machine learning, and related research and development with industries.

M4: To provide an enjoyable environment for pursuing excellence while upholding strong personal and professional values and ethics.

PROGRAM EDUCATIONAL OBJECTIVES (PEOS)

PEO 1: Excel in technical abilities to build intelligent systems in the fields of AI & ML in order to find new opportunities.

PEO 2: Embrace new technology to solve real-world problems, whether alone or as a team, while prioritizing ethics and societal benefits.

PEO 3: Accept lifelong learning to expand future opportunities in research and product development.

PROGRAM OUTCOMES

Engineering students will be able to:

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: Expertise in tailoring ML algorithms and models to excel in designated applications and fields.

PSO2: Ability to conduct research, contributing to machine learning advancements and innovations that tackle emerging societal challenges.

ABSTRACT

Artificial Intelligence (AI) tools such as ChatGPT, GitHub Copilot, Gemini AI, Microsoft Edge, Meta AI, Claude, Perplexity AI, and Quillbot are increasingly becoming part of modern classrooms, supporting students in tasks like coding, writing, grammar checking, problem-solving, and idea generation. While these tools offer significant academic benefits, their actual influence on student learning, engagement, and responsible usage remains largely unmeasured. Most institutions still lack a structured system to capture and analyze student experiences with AI tools, creating uncertainty for educators regarding their effectiveness and potential risks. This project introduces an AI Tools Feedback Tracker System, designed to systematically collect, preprocess, and analyze student reflections on AI tool usage. Student feedback is gathered through online surveys and processed using IBM Watson Studio, where the dataset undergoes cleaning, formatting, and validation to ensure accuracy. The cleaned data is then visualized using Watson's analytical tools and further represented in an interactive dashboard created in RStudio using Shiny. These visualizations help identify the most frequently used tools, overall satisfaction levels, common challenges, and emerging trends. The system not only provides insights into positive and negative student experiences but also supports educators in recognizing issues such as misuse, over-dependence, or misunderstandings related to AI tools. By offering data-driven reports, visual dashboards, and categorized feedback analysis, the project empowers institutions to make informed decisions about responsible AI integration in education. Ultimately, the Feedback Tracker enhances transparency, improves learning outcomes, and supports the creation of effective policies for ethical and productive AI adoption in classrooms.

ABSTRACT WITH POs AND PSOs MAPPING

ABSTRACT	POs MAPPED	PSOs MAPPED
AI tools like ChatGPT, GitHub Copilot, Gemini AI, Microsoft Edge, Meta AI, Claude, Perplexity AI and Quillbot are rapidly entering classrooms, offering assistance in coding, writing, and learning. However, their actual impact on students is rarely reviewed or tracked. This project proposes a Feedback Tracker System that collects student reflections on AI tools and generates insights for teachers and administrators. The system ensures responsible usage, identifies benefits and challenges, and helps institutions make data-driven decisions on AI integration in education.	PO1(3) PO2(3) PO3(3) PO4(2) PO5(3) PO6(2) PO7(2) PO8(1) PO9(2) PO10(2) PO10(3) PO11(2) PO12(2)	PSO1(1) PSO2(2)

Note: 1- Low, 2-Medium, 3- High

SUPERVISOR

HEAD OF THE DEPARTMENT

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1.INTRODUCTION

1.1 Project Objective

The objective of this project is to collect, analyze, and visualize student feedback on various AI tools used in classrooms. It aims to identify the most commonly used tools, understand student satisfaction levels, and observe learning outcomes based on their reflections. By processing the collected data and generating meaningful insights, the project helps teachers and educational institutions track AI usage trends, detect challenges such as misuse or over-dependence, and make data-driven decisions for responsible AI integration in education.

1.2 Technologies Used

1. Data Collection

- Google Forms / Online Survey Platform
- Used to collect student feedback on various AI tools.

2.Data Cleaning & Preprocessing

- **IBM Watson Studio**
 - Removing duplicates and missing values
 - Standardizing column names
 - Formatting ratings
 - Preparing final cleaned dataset

3. Data Visualization & Analysis

- **IBM Watson Studio (Built-in Visual Analytics Tools)**
 - Bar charts, pie charts, frequency graphs
 - Tool-wise usage comparison
 - Rating distribution visualization

4. Dashboard Development

- **RStudio (Shiny / ggplot2 / dplyr)**
- Used to build an interactive dashboard showing:
 - Tool usage
 - Ratings
 - Feedback insights
 - Summary statistics

5. File Storage & Data Handling

- CSV Format / Microsoft Excel
- Used for storing raw and cleaned datasets before importing into tools.

2. PROJECT REQUIREMENTS

2.1 Input Data Description

The input data consists of student feedback collected through an online form to understand their usage and experience with different AI tools in the classroom. After collecting the responses, the dataset was cleaned and prepared using IBM Watson Studio. The cleaned dataset contains the following columns:

1. Name

Represents the student's name or identifier.

- Used only for reference/identification.
- Optional in visualization (can be removed for privacy).

Example:

"Viveka", "Arun", "Meena"

2. AI_Tool

- Indicates the AI tool used by the student.
- Categorical (non-numerical) column.
- Used to analyze which tools are most popular.

Examples:

- ChatGPT
- GitHub Copilot
- Gemini
- Grammarly
- Quillbot

3. Feedback / Experience

- A text-based comment where students describe their experience.
- Used for reading opinions, understanding issues, or sentiment analysis.
- Cleaned during preprocessing to remove empty or invalid entries.

Example:

"The tool was very helpful for coding practice."

"It sometimes gives slow responses."

4. Rating

- A numeric value indicating the student's satisfaction level.
- Converted into numbers during data cleaning (1–5 scale).

Scale:

- 1 – Very Poor
- 2 – Poor
- 3 – Average
- 4 – Good
- 5 – Excellent

Used to create rating distribution charts.

2.2 Expected Output

1. AI Tool Usage Comparison Chart

– A bar chart showing how many students used each AI tool (ChatGPT, Copilot, Gemini, Quillbot, etc.).

2. Overall Rating Distribution Chart

– A visualization displaying how students rated AI tools on a scale of 1–5.

3. Tool-Wise Rating Comparison

– Side-by-side charts comparing the average rating of each AI tool to identify the most effective platform.

4. Feedback Trend Analysis

– A chart showing improvement, decline, or consistency in students' feedback over time (if timestamp data is available).

5. Interactive Feedback Table

– A searchable and filter-enabled table showing all student details, tool names, and feedback comments for quick review.

6. Data Summary Panel

– Automatic display of total responses, number of tools used, highest-rated AI tool, and common feedback patterns.

7. User-Friendly Dashboard Interface

– A multi-tab dashboard built in R Shiny with upload options, visualizations, summary panels, and a clean layout for teachers and administrators.

3.DESIGN AND ARCHITECTURE

3.1 Frontend Overview

The frontend is the user-facing interface where students submit their feedback about various AI tools. It is designed to be simple, responsive, and easy to use. The interface is typically created using Google Forms or a web-based form, allowing students to enter details such as the AI tool they used, their feedback, and a rating.

The frontend ensures:

- Smooth data entry
- Mobile-friendly design
- Clean layout for easy navigation
- A structured format that reduces errors during submission

All collected data is automatically exported in CSV format, ready for preprocessing in IBM Watson Studio.

3.2 Backend Overview

The backend processes, cleans, and prepares the collected feedback for analysis and dashboard visualization. In this project, backend operations are performed using IBM Watson Studio and RStudio.

In IBM Watson Studio:

- Imported the raw feedback CSV file
- Removed duplicates and blank rows
- Standardized tool names and column formatting
- Extracted numeric rating values
- Handled missing values
- Saved the cleaned dataset for analysis

In RStudio:

- Loaded the cleaned dataset
- Applied data manipulation using libraries like dplyr
- Created visual outputs using ggplot2
- Built an interactive dashboard using R Shiny

The backend ensures that only valid, high-quality, structured data is used for visualization and insights.

3.3 Interaction Flow

AI Tools Feedback

* Required

Name *

AI Tool Used *

Select AI Tool

Feedback / Experience

Rating

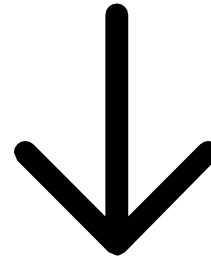
☆
☆
☆
☆

Submit

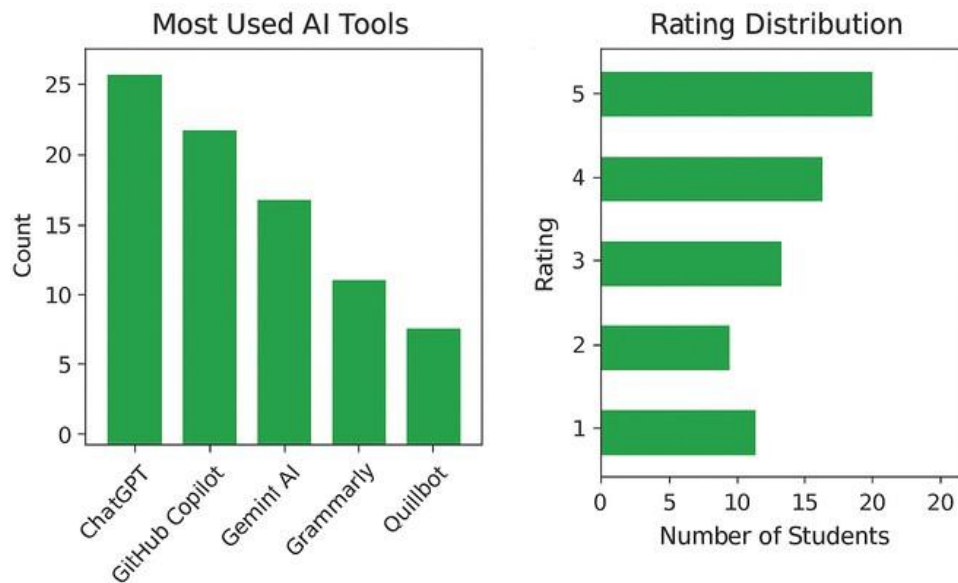
→

Name	AI Tool	Feedback	Rating
Viveka	ChatGPT	Very usefu	5
Vannika	GitHub Co	Helps with	4
Kaviya	Gemini	Good for g	4
Deva	Grammarly	Helps corr	5
Lakshana	Quillbot	Useful for i	3
Mirthika	ChatGPT	Very usefu	5
Darshan	GitHub Co	Helps with	4
Arun	Gemini	Good for g	4
Karthik	Grammarly	Helps corr	5
Renu	Quillbot	Useful for i	3
Anu	ChatGPT	Very usefu	5
Darshini	GitHub Co	Helps with	4
Monica	Gemini	Good for g	4
Giruja	Grammarly	Helps corr	5
Nandhini	Quillbot	Useful for i	3
Priya	ChatGPT	Very usefu	5
Charu	GitHub Co	Helps with	4
Aadhiksha	Gemini	Good for g	4
Bhavana	Grammarly	Helps corr	5
Chindana	Quillbot	Useful for i	3
Bharathi	ChatGPT	Very usefu	5

AI Tools_Feedback_Dataset



AI Tools Usage-Feedback Tracker in Classroom

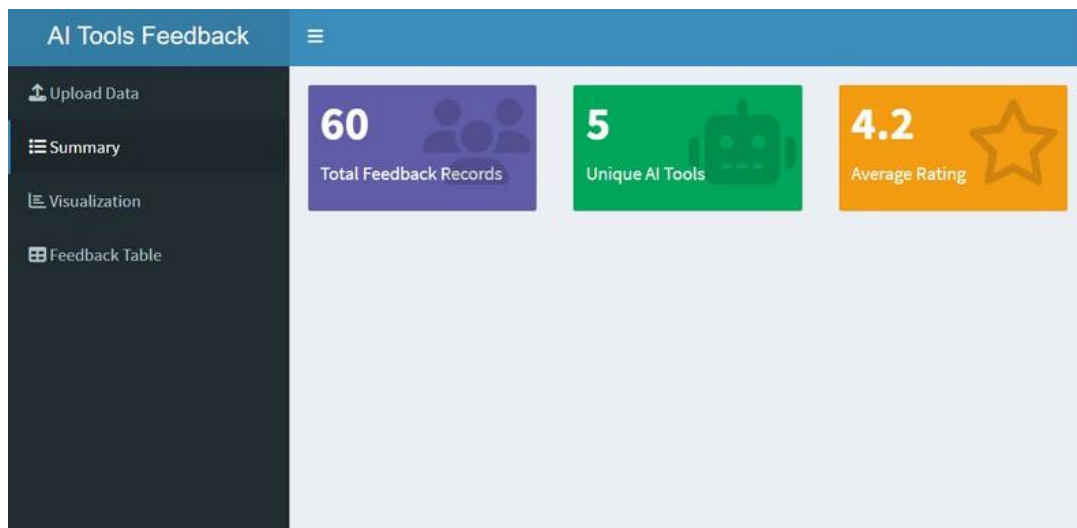


4.IMPLEMENTATION

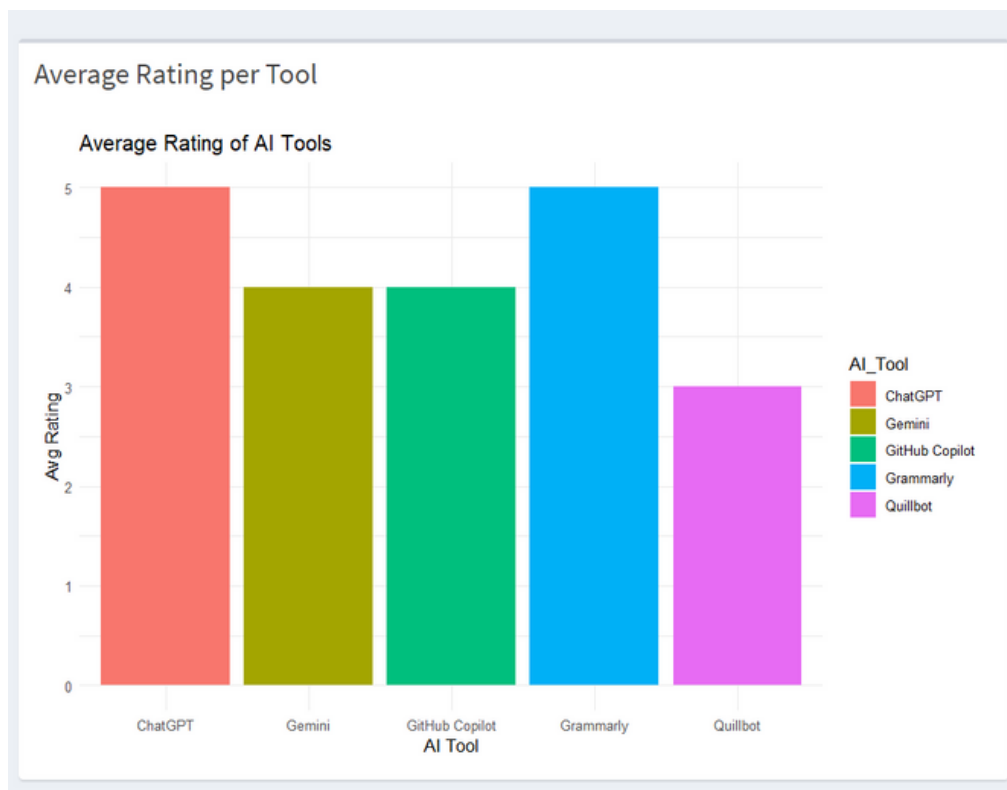
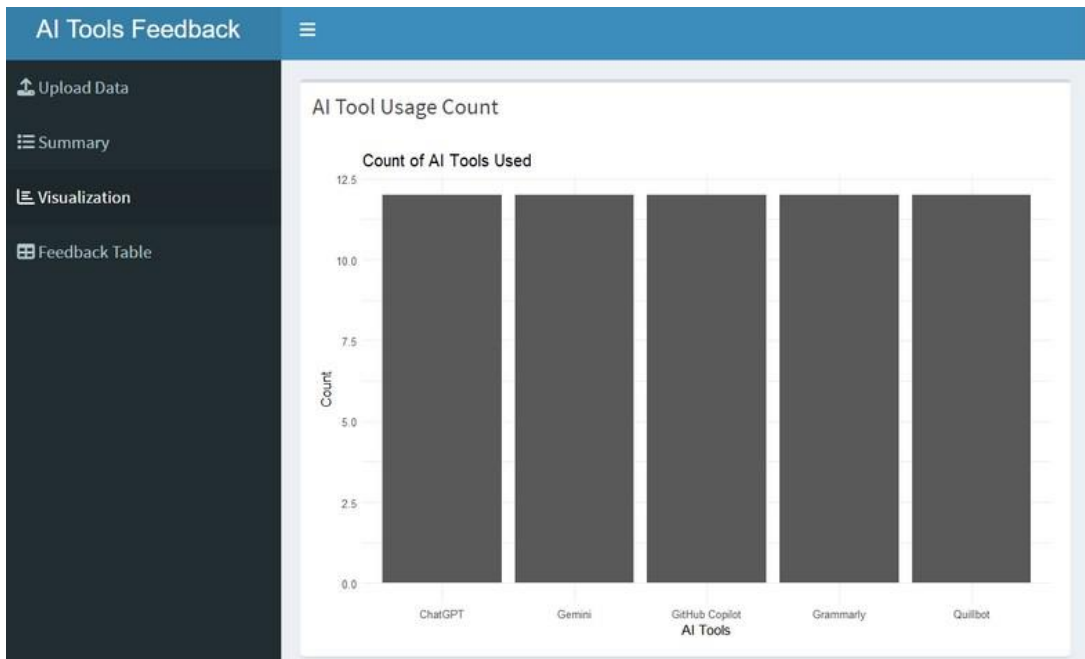
4.1 DATASET

A	B	C	D	E	F	G	H	I	J	K	L	M	N
Name	AI_Tool	Feedback	Rating										
Viveka	ChatGPT	Very usefu	5										
Vannika	GitHub Co	Helps with	4										
Kaviya	Gemini	Good for g	4										
Deva	Grammarly	Helps corr	5										
Lakshana	Quillbot	Useful for i	3										
Mirthika	ChatGPT	Very usefu	5										
Darshan	GitHub Co	Helps with	4										
Arun	Gemini	Good for g	4										
Karthik	Grammarly	Helps corr	5										
Renu	Quillbot	Useful for i	3										
Anu	ChatGPT	Very usefu	5										
Darshini	GitHub Co	Helps with	4										
Monica	Gemini	Good for g	4										
Giruja	Grammarly	Helps corr	5										
Nandhini	Quillbot	Useful for i	3										
Priya	ChatGPT	Very usefu	5										
Charu	GitHub Co	Helps with	4										
Aadhiksha	Gemini	Good for g	4										
Bhuvana	Grammarly	Helps corr	5										
Chindana	Quillbot	Useful for i	3										
Bharathi	ChatGPT	Very usefu	5										

4.2 DASHBOARD



5.DATA VISUALIZATIONN



5.1 SUMMARY TABLE

AI Tools Feedback				
Upload Data Summary Visualization Feedback Table All Feedback Show 10 entries Search:				
	Name	AI_Tool	Feedback	Rating
1	Viveka	ChatGPT	Very useful for learning.	5
2	Vannika	GitHub Copilot	Helps with coding tasks.	4
3	Kaviya	Gemini	Good for generating ideas.	4
4	Deva	Grammarly	Helps correct grammar mistakes.	5
5	Lakshana	Quillbot	Useful for rewriting content.	3
6	Mirthika	ChatGPT	Very useful for learning.	5
7	Darshan	GitHub Copilot	Helps with coding tasks.	4
8	Arun	Gemini	Good for generating ideas.	4
9	Karthik	Grammarly	Helps correct grammar mistakes.	5
10	Renu	Quillbot	Useful for rewriting content.	3

6. CHALLENGES AND SOLUTIONS

6.1 Issues Encountered

1. Inconsistent and Incorrect Student Responses:

Some students entered tool names with spelling mistakes (Ex: “Chatgpt”, “copilot”), provided incomplete feedback, or left blank fields.

2. Mixed Rating Formats:

The rating column contained values like “4 – Good”, “Five”, or text descriptions instead of numeric values, making preprocessing difficult.

3. Duplicate and Empty Records:

Multiple repeated entries and blank responses were found in the dataset and had to be removed manually.

4. Data Type Conversion Errors:

Converting ratings and textual feedback into correct numeric and categorical types caused validation issues.

5. Performance Lag in R Shiny Dashboard:

Rendering charts and tables took time depending on system performance, especially with larger datasets.

6. Teacher Interpretation Issues:

Some teachers found it difficult to interpret bar charts, rating distributions, and sentiment patterns without guidance.

7. Student Honesty Problem:

A few responses seemed unrealistic or biased since the system relied on truthful reflections.

6.2 Resolution Strategies

1. Standardizing Data Collection:

To avoid inconsistent or incorrect student responses, dropdown menus and required fields were added in the Google Form. This ensured uniform tool names, valid ratings, and complete entries.

2. Rating Normalization:

All mixed-format ratings (e.g., “4 – Good”, “five”, “3”) were converted into clean numeric values using IBM Watson Studio’s transformation functions. This allowed proper statistical analysis and visualization.

3. Data Cleaning Automation:

Duplicate, blank, or irrelevant entries were removed using automated cleaning tools in Watson Studio. Unnecessary columns generated by Google Forms were deleted to keep the dataset focused.

4. Data Type Validation:

Columns such as rating, tool name, and feedback were validated and converted into correct data types (numeric / categorical / text) to avoid processing issues and graph errors.

5. Improving Visualization Reliability:

Missing values were handled, and column names were standardized so that charts in Watson Studio and R Shiny loaded without misalignment or blank outputs.

6. Teacher Guidance & Simple UI:

A simple, intuitive dashboard interface was created so that teachers could easily understand charts. Tooltips, labels, and descriptions were added for clarity.

7. Monitoring for Honest Responses:

Reflection-based forms and open-ended questions were used to encourage honest student feedback. Irregular responses were flagged for review.

CONCLUSION

The AI Tools Feedback Tracker successfully provides a structured and efficient way to collect, clean, and analyze student reflections on AI tools used in the classroom. By using Google Forms for data collection, IBM Watson Studio for preprocessing and visualization, and RStudio Shiny for interactive dashboards, the project offers a complete end-to-end solution for understanding AI usage trends among students. The system highlights the most frequently used tools, student satisfaction levels, and common challenges, enabling educators to make informed, data-driven decisions. It also supports responsible AI usage by identifying patterns of over-dependence, potential misuse, or ineffective tool adoption. Overall, this project enhances transparency, improves learning insights, and promotes meaningful integration of AI in educational environments.

FUTURE SCOPE

1. Integrate real-time data syncing for live feedback updates.
2. Add advanced sentiment analysis using AI/ML models.
3. Develop a mobile app for easier feedback submission.
4. Implement login-based personalized dashboards for students and teachers.
5. Connect with LMS platforms like Google Classroom or Moodle.
6. Enable automatic monthly/annual report generation.

REFERENCES

1. IBM Watson Studio Documentation. IBM Cloud.
2. RStudio Team. RStudio: Integrated Development Environment for R. RStudio, PBC.
3. Chang, W., Cheng, J., Allaire, J., Xie, Y., & McPherson, J. (2021). Shiny: Web Application Framework for R.
4. Google Forms Help Center. Create & Analyze Surveys.

APPENDIX

```
library(shiny)
library(ggplot2)
library(dplyr)
library(shinydashboard)

ui <- dashboardPage(

  dashboardHeader(title = "AI Tools Feedback Dashboard"),

  dashboardSidebar(
    sidebarMenu(
      menuItem("Upload Data", tabName = "upload", icon = icon("upload")),
      menuItem("Summary", tabName = "summary", icon = icon("list")),
      menuItem("Visualization", tabName = "visual", icon = icon("chart-bar")),
      menuItem("Feedback Table", tabName = "table", icon = icon("table"))
    )
  ),

  dashboardBody(

    tabItems(

      # PAGE 1 – Upload Section
      tabItem(tabName = "upload",
        fluidRow(
          box(width = 6, fileInput("file", "Upload CSV File", accept = ".csv")),
          box(width = 6, title = "Instructions",
            "Upload your cleaned dataset (Feedback Tracker CSV)")
        )
      ),

      # PAGE 2 – Summary
      tabItem(tabName = "summary",
        fluidRow(
          valueBoxOutput("total_records"),
          valueBoxOutput("unique_tools"),
          valueBoxOutput("avg_rating")
        )
      ),

    )
  )
)
```

PAGE 3 – Visualization

```
tabItem(tabName = "visual",  
  fluidRow(  
    box(width = 12, title = "AI Tool Usage Count",  
      plotOutput("tool_plot"))  
  ),  
  
  fluidRow(  
    box(width = 12, title = "Average Rating per Tool",  
      plotOutput("rating_plot"))  
  )  
)
```

PAGE 4 – Feedback Table

```
tabItem(tabName = "table",  
  fluidRow(  
    box(width = 12, title = "All Feedback",  
      dataTableOutput("feedback_table"))  
  )  
)  
)  
)  
)
```

```
server <- function(input, output) {
```

```
  # READ DATA
```

```
  data <- reactive({  
    req(input$file)  
    read.csv(input$file$datapath)  
  })
```

```
  # SUMMARY BOXES
```

```
  output$total_records <- renderValueBox({  
    req(data())  
    valueBox(nrow(data()), "Total Feedback Records", icon = icon("users"), color = "purple")  
  })
```

```
output$unique_tools <- renderValueBox({
  req(data())
  valueBox(length(unique(data())$AI_Tool), "Unique AI Tools", icon = icon("robot"), color =
"green")
})

output$avg_rating <- renderValueBox({
  req(data())
  valueBox(round(mean(data())$Rating), 2), "Average Rating", icon = icon("star"), color =
"yellow")
})

# TOOL COUNT BAR CHART
output$tool_plot <- renderPlot({
  req(data())
  ggplot(data(), aes(AI_Tool)) +
    geom_bar() +
    theme_minimal() +
    labs(title = "Count of AI Tools Used",
         x = "AI Tools", y = "Count")
})

# AVERAGE RATING CHART
output$rating_plot <- renderPlot({
  req(data())
  df <- data() %>% group_by(AI_Tool) %>% summarise(Average_Rating = mean(Rating))

  ggplot(df, aes(AI_Tool, Average_Rating, fill = AI_Tool)) +
    geom_col() +
    theme_minimal() +
    labs(title = "Average Rating of AI Tools",
         x = "AI Tool", y = "Avg Rating")
})

# FEEDBACK TABLE
output$feedback_table <- renderDataTable({
  req(data())
  data()
})
}

shinyApp(ui = ui, server = server)
```